in a separate tank. After mating, the eggs are laid. Spawning tanks of different sizes should be prepared with proper aeration. Temperature (24^0 C – 30^0 C) and pH (7-8) should be maintained in the hatching tank. The eggs hatch into first and second stage larva. Artificial feed is supplied. Young ones of 5cm length (60 days old) can be reared in fresh or slightly brackish water ponds and paddy fields. Harvesting of prawns can be done twice in a year.



Figure 12.10 *Macrobrachium rosenbergii*

Culture of marine prawn

Several factors that determine the success of marine prawn culture includes selection of site, water quality, soil quality and availability of seed.

Preparation of farm

For the preparation of ponds for algal growth and for the subsequent stocking of prawns it is essential to drain off the water and sundry the bottom followed by light tilling. Agricultural lime should be applied to absorb excess CO_2 and to supply calcium which is required for moulting. Fertilizers like rice, bran, poultry, and cattle dung are used to increase the fertility of the soil. Prawns are commonly caught in crafts and gears using different types of nets such as cast nets, bag nets, drag nets, trawl nets and barrier nets. Preservation of prawns is done by peeling and deveining or by cooking and peeling.

12.7.3 Pearl Culture

Pearl is a white, highly shining globular concretion found within the shell of an Oyster. Pearl oysters are sedentary animals. In India it was cultured for the first time in 1973 at Thoothukudi. Pearl oysters are found along the coast of Kanyakumari and in the Gulf of Kutch. High quality pearls are obtained from pearl oysters of Genus *Pinctada* that can be cultured in the salinity range of 30 ppt in racks, raft and long line methods. Freshwater bivalve *Lamellidens* is also used in artificial pearl culture. Mostly the pearl oysters inhabit the ridges of rocks or dead corals, forming extensive pearl banks. These pearl beds produce best quality of pearls called as "Lingha Pearl".

Pearl Formation

When a foreign particle accidently enters into the space between mantle and shell of the oyster, it adheres to the mantle. The mantle epithelium encloses it like a sac and starts to secrete concentric layers of nacre around it as a defensive mechanism. Nacre is secreted continuously by the epithelial layer of the mantle and is deposited around the foreign particle and over a period time the formation of repeated layers of calcium carbonate makes the hard and glossy pearl. When the pearl enlarges the oyster dies. The shell is then carefully opened and the pearls are manually separated and graded (Figure 12. 11).

Composition of pearl

Pearl comprises of water, organic matter, calcium carbonate and the residue.

(1) Water: 2-4% (2) Organic matter: 3.5-5.9%

(3) Calcium carbonate: 90% (4) Residue: 0.1-0.8%

TN_GOVT_XI_Bio Zoology_Ch12.indd 232 09-01-2020 5.33.20 PM